

Week 10 – Workshop

Geothermics: Finite Differences



Geophysics 3A: Potential Fields & Geothermics

July 6, 2026

Duration: 3 hours

Setting: Workshop room

Preparation

Pre-reading:

Lecture videos:

Notes:

Purpose

This workshop builds intuition for finite difference methods through physical reasoning and interactive activities. Students first derive the heat equation from energy conservation, then experience how numerical models work by acting as nodes that update temperature based on their neighbours. The workshop then explores how boundary conditions represent geological assumptions and how numerical stability reflects limits on how heat can physically propagate. By the end, students should understand that finite difference methods are simply local energy balance, that heat flow emerges from neighbour interactions, and that both boundary conditions and stability have direct physical meaning.

Learning Outcomes

Students will:

- Understand finite difference equations as local statements of energy conservation.
- Develop intuition for how numerical solutions evolve through neighbour interactions.
- Interpret boundary conditions as geological assumptions.
- Understand numerical stability as a limit on physical information transfer.

Session Flow (180 min)

0–30 min: Activity A: From Conservation of Energy to Finite Differences

Concept

Finite difference equations arise from applying conservation of energy to small rock volumes.

- Nodes represent local rock volumes.
- Heat flux enters and leaves each node.
- Heat production adds energy locally.

The equation is *not a numerical trick*. The Laplacian as a measure of curvature (difference from neighbours). This is just bookkeeping of energy.

30–65 min: Activity B: Finite Differences as a Physical Process: Human Finite Difference Solver**Concept**

Numerical models evolve through local interactions between neighbouring nodes.

Discuss:

- smoothing of temperature
- spreading of heat
- role of gradients

65–85 min: Activity C: Boundary Conditions as Geological Assumptions

Concept: Boundary conditions determine what physical system the model represents.

- Fixed temperature (Dirichlet)
- Fixed heat flux (Neumann)

85–95 min: Break**95–120 min: When Analytic Solutions Break Down****Concept**

- Analytic solutions require simplifications:
 - constant properties
 - simple geometries
 - simple boundary conditions
- Real geology violates these assumptions.

Discussion prompts

- What makes geological systems difficult to solve analytically?
- Why do irregular boundaries force numerical approaches?

Your Iberia exercise on the problem has several issues that make it impossible to develop an analytic form, including variable sediment thickness, lateral structure, varying boundary condition.

120–160 min: Activity D: Numerical Stability and Information Transfer

Concept Stability limits impose a “speed limit” on thermal information:

$$\Delta t < \frac{\Delta x^2}{2\kappa}$$

- Heat propagates gradually through diffusion.
- Numerical schemes must respect this.
- small vs large time steps
- what happens when updates are too large
- stability = causality
- information cannot skip nodes

Instability is not just numerical—it is physically impossible behaviour.

160–180 min: Wrap-Up and Synthesis**Key connections**

- Finite differences = local energy balance + neighbour interaction
- Boundary conditions = geological assumptions
- Stability = physical constraint on information transfer

Discussion

- How does the numerical method compare to analytic solutions?
- What advantages do numerical models provide?
- Where might these methods be applied beyond heat flow?

Preview

- Extension to 2-D systems
- Application to real datasets (Iberia Abyssal Plain)

Activity A

From Conservation of Energy to Finite Differences

Purpose

To understand finite difference heat flow equations as direct consequences of energy conservation applied to small volumes of rock.

Consider a vertical column of sediment divided into small layers. Heat flows between layers, and each layer can gain or lose energy depending on its neighbours and any internal heat production.

Rather than solving a continuous equation, we will track how heat moves between *discrete points*.

Part A: Physical intuition

- If a layer is hotter than the ones above and below it, does it gain or lose heat?
- If heat is being produced inside the layer, what happens over time?

Part B: Setting up the system

- Sketch a one-dimensional grid of nodes labeled T_{i-1} , T_i , and T_{i+1} .
- Indicate the spacing Δz .

Part C: Heat flux

Heat flows according to Fourier's law:

$$q_z = -k \frac{\partial T}{\partial z} \quad (1)$$

Tasks:

- Write an expression for heat flowing *into* node i .
- Write an expression for heat flowing *out of* node i .

Part D: Energy conservation

Energy conservation for a small volume gives:

$$\text{Rate of change} = \text{in} - \text{out} + \text{produced}$$

Task: Combine your expressions to derive:

$$\frac{\partial T_i}{\partial t} = \kappa \frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta z^2} + \frac{A_i}{\rho c_p} \quad (2)$$

Part E: Interpretation

Explain in words:

- What does the term $T_{i+1} - 2T_i + T_{i-1}$ represent?

- What role does heat production A_i play?
- What happens if $A_i = 0$?

Key Insight

Finite difference equations are not numerical tricks—they are simply energy conservation applied locally.

Activity B

Solving the Heat Equation as a Human Computer

Purpose

To develop intuition for finite difference methods by acting as the nodes in a numerical heat flow model.

In a numerical model, each point (node) updates its temperature using only information from its neighbours.

In this activity, you will act as the nodes in a finite difference model.

Box .1: Human Computer

Before electronic computers existed, a *computer* was literally a person employed to carry out calculations by hand or with mechanical calculators. Large groups of human computers were used in astronomy, navigation, ballistics, engineering, and later for the numerical solution of differential equations. By the late 19th and early 20th centuries, many of these computing groups were composed largely of women. Examples include the astronomical computers working for the *Nautical Almanac* and later teams at observatories, government laboratories, and research institutions.

During World War II, the scale of numerical work *exploded* (pun intended). Hundreds of human computers computed ballistic firing tables and solved systems of finite-difference equations for artillery trajectories, fluid flow, and other military problems. Post-WWII, human computers were also instrumental to the US space program. If you've seen the movie *Hidden Figures*, you learned about some of these remarkable women who helped make going to the Moon possible by producing many of the calculations needed to determine flight plans, rocket burn parameters, launch windows, and safe re-entry trajectories.

Finite-difference methods were especially important because many physical problems could not be solved analytically. Human computers discretized differential equations onto grids and repeatedly computed numerical updates row by row—an extremely laborious process that helped motivate the development of electronic computers. The first six programmers of the world's first general-purpose electronic computer, ENIAC, were women who had previously worked as human computers.

David Alan Grier, **2005**, *When Computers Were Human*, Princeton University Press.

Kathy Kleiman, **2022**, *Proving Ground: The Untold Story of the Six Women Who Programmed the World's First Modern Computer*.

David Alan Grier, **2001**, *Human computers: the first pioneers of the information age*, **Endeavour** 25(1).

Setup

- Each student is assigned a position in a line.
- Each student is given an initial temperature value T_i .
- The two students at the ends of the line represent **fixed boundary conditions**.

Update rule

At each step, interior nodes update their temperature according to:

$$T_i^{\text{new}} = \frac{T_{i-1} + T_{i+1}}{2} \quad (3)$$

This means:

- you look at your two neighbours,
- take the average,
- and update your value.

Part A: Prediction

Before starting, discuss:

- What happens to a sharp temperature spike over time?
- Will the system become smoother or less smooth?

Part B: Iteration

Repeat the following for several steps:

1. Show your value to your neighbours.
2. Compute your new value.
3. Update simultaneously.

Record the evolution of the system.

Part C: Observations

Discuss:

- Where do the largest changes occur?
- What happens to temperature peaks?
- How does heat move through the system?

Part D: Interpretation

Explain in words:

- Why does the system become smoother over time?
- What role do neighbouring nodes play?
- How does this relate to diffusion?

Key Insight

Heat flow is a local process: each point evolves based on its neighbours.

Optional Extension 1: Heat Production

Introduce a constant heat source at one node.

New update rule:

$$T_i^{\text{new}} = \frac{T_{i-1} + T_{i+1}}{2} + S \quad (4)$$

where S is a constant heat input.

Tasks:

- What happens to the system over time?
 - Does it still become uniform?
 - What steady-state behaviour emerges?
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Optional Extension 2: Two-Dimensional Grid

Arrange students in a grid.

Update rule:

$$T_{i,j}^{\text{new}} = \frac{T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}}{4} \quad (5)$$

Tasks:

- Observe how a hot region spreads in two dimensions.
- Compare to the one-dimensional case.
- Describe the shape of the temperature field over time.

Final Reflection

- In what sense are you “solving the heat equation”?
- Why does this method require many iterations?

Activity C

Boundary Conditions and Geological Meaning

Purpose

To understand how boundary conditions encode geological assumptions in numerical models.

A numerical model does not “know” geology—it only knows equations and boundary conditions.

Boundary conditions determine what kind of Earth your model represents.

Part A: Grounding in real systems

Consider the following settings:

- Ocean floor beneath deep water
- Continental lithosphere
- A hot magma body cooling in the crust

Task: For each case, describe what controls the temperature at the boundaries.

Part B: Types of boundary conditions

Common thermal boundary conditions:

- Fixed temperature (Dirichlet)
- Fixed heat flux (Neumann)

Task: Match each type of boundary condition to a geological situation.

Part C: Comparing assumptions

Explain how the model changes if you prescribe:

- surface temperature
- surface heat flow

What is fixed in each case? What is allowed to vary?

Part D: Geological consequences

Compare appropriate boundary conditions for:

- oceanic lithosphere
- continental lithosphere

Part E: Critical thinking

Task: Explain why inappropriate boundary conditions can produce physically misleading results—even if the numerical solution is mathematically stable.

Reflection

If two models use the same equation but different boundary conditions, can they produce completely different results? Why?

Key Insight

Boundary conditions are where geological assumptions enter a numerical model.

Activity D

Stability, Time Stepping, and Information Transfer

Purpose

To develop physical intuition for numerical stability in explicit finite difference schemes.

In a time-stepping model, temperature is updated step-by-step. Each step uses information from neighbouring nodes.

Think of heat as spreading gradually through the system.

Part A: Prediction

Consider two cases:

- very small time steps
- very large time steps

What do you expect to happen in each case?

Part B: Physical reasoning

Task: Explain why heat cannot move arbitrarily far in a single time step.

What determines how quickly heat diffuses through a material?

Part C: Stability condition

For a one-dimensional explicit scheme:

$$\Delta t < \frac{\Delta z^2}{2\kappa} \quad (6)$$

Tasks:

- Why does the allowable time step depend on Δz ?
- Why does it depend on thermal diffusivity κ ?

Part D: Interpretation

Task: Interpret the stability condition as a limit on how fast thermal information can move between nodes.

Why can heat not “jump” over nodes in a single time step?

Part E: Geological interpretation

Explain:

- Why are sedimentary basins (low κ) easier to model?
- Why do high-diffusivity materials require smaller time steps?

Key Insight

Numerical instability occurs when the model allows heat to propagate faster than physically possible.